

```
MariaDB [(none)]> create database work;  
Query OK, 1 row affected (0.003 sec)
```

```
MariaDB [(none)]> use work;  
Database changed
```

Question 7.24

```
MariaDB [work]> CREATE TABLE Department (deptNo CHAR(4) PRIMARY KEY,deptName VARCHAR(50) NOT  
NULL,mgrEmpNo CHAR(5));  
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [work]> CREATE TABLE Employee ( empNo CHAR(5) PRIMARY KEY, fName VARCHAR(30) NOT NULL, lName  
VARCHAR(30) NOT NULL, address VARCHAR(100), DOB DATE, sex CHAR(1) NOT NULL, position VARCHAR(30) NOT  
NULL, deptNo CHAR(4), FOREIGN KEY (deptNo) REFERENCES Department(deptNo), CONSTRAINT chk_Employee CHECK  
(sex IN ('M', 'F') AND position IN ('Manager', 'Team Leader', 'Analyst', 'Software Developer')));  
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [work]> ALTER TABLE Department ADD FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo);  
Query OK, 0 rows affected (0.020 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> CREATE TABLE Project ( projNo CHAR(5) PRIMARY KEY, projName VARCHAR(50) NOT NULL, deptNo  
CHAR(4) NOT NULL, FOREIGN KEY (deptNo) REFERENCES Department(deptNo));  
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [work]> CREATE TABLE WorksOn ( empNo CHAR(5) NOT NULL, projNo CHAR(5) NOT NULL, dateWorked DATE  
NOT NULL, hoursWorked INT NOT NULL, PRIMARY KEY (empNo, projNo, dateWorked), FOREIGN KEY (empNo)  
REFERENCES Employee(empNo), FOREIGN KEY (projNo) REFERENCES Project(projNo), CONSTRAINT chk_Hours CHECK  
(hoursWorked BETWEEN 0 AND 40));  
Query OK, 0 rows affected (0.010 sec)
```

-- Let's populate our database

```
MariaDB [work]> INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES ('D001', 'IT and  
Infrastructure', NULL), ('D002', 'Planning & Strategy', NULL), ('D003', 'Human Resources', NULL),  
( 'D004', 'Finance & Accounting', NULL), ('D005', 'Research & Development', NULL), ('D006', 'Marketing &  
Sales', NULL), ('D007', 'Customer Success', NULL), ('D008', 'Legal & Compliance', NULL), ('D009',  
'Quality Assurance', NULL), ('D010', 'Operations', NULL);  
Query OK, 10 rows affected (0.006 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> INSERT INTO Employee (empNo, fName, lName, address, DOB, sex, position, deptNo) VALUES  
( 'E0001', 'Alice', 'Smith', '123 Maple St', '1985-04-12', 'F', 'Manager', 'D001'), ('E0002', 'Bob',  
'Jones', '456 Oak Ave', '1992-08-23', 'M', 'Software Developer', 'D001'), ('E0003', 'Carol', 'Taylor',  
'789 Pine Rd', '1988-11-02', 'F', 'Team Leader', 'D002'), ('E0004', 'David', 'Brown', '101 Cedar Ln',  
'1995-01-15', 'M', 'Analyst', 'D002'), ('E0005', 'Emma', 'Davis', '202 Birch Dr', '1991-06-30', 'F',  
'Manager', 'D005'), ('E0006', 'Frank', 'Miller', '303 Elm St', '1983-09-18', 'M', 'Team Leader', 'D005'),  
( 'E0007', 'Grace', 'Wilson', '404 Walnut Av', '1994-03-22', 'F', 'Software Developer', 'D005'), ('E0008',  
'Henry', 'Moore', '505 Ash Rd', '1979-12-05', 'M', 'Manager', 'D004'), ('E0009', 'Ivy', 'Thomas', '606  
Cherry Ln', '1996-07-14', 'F', 'Analyst', 'D001'), ('E0010', 'Jack', 'Anderson', '707 Plum St',  
'1990-10-25', 'M', 'Software Developer', 'D009');  
Query OK, 10 rows affected (0.005 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

-- Give the departments their managers

```
MariaDB [work]> UPDATE Department SET mgrEmpNo = 'E0001' WHERE deptNo = 'D001';  
Query OK, 1 row affected (0.009 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
MariaDB [work]> UPDATE Department SET mgrEmpNo = 'E0003' WHERE deptNo = 'D002';  
Query OK, 1 row affected (0.001 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
MariaDB [work]> UPDATE Department SET mgrEmpNo = 'E0005' WHERE deptNo = 'D005';  
Query OK, 1 row affected (0.001 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
MariaDB [work]> UPDATE Department SET mgrEmpNo = 'E0008' WHERE deptNo = 'D004';  
Query OK, 1 row affected (0.002 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [work]> INSERT INTO Project (projNo, projName, deptNo) VALUES ('P0001', 'Cloud Migration',  
'D001'), ('SCCS', 'Secure Core Control Sys', 'D001'), ('P0003', 'Q3 Strategic Mapping', 'D002'),  
( 'P0004', 'HR Portal Automation', 'D003'), ('P0005', 'AI Predictive Engine', 'D005'), ('P0006', 'ERP  
Ledger Integration', 'D004'), ('P0007', 'Global Ad Campaign', 'D006'), ('P0008', 'CRM System Overhaul',  
'D007'), ('P0009', 'GDPR Compliance Audit', 'D008'), ('P0010', 'Automated QA Pipelines', 'D009');  
Query OK, 10 rows affected (0.001 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES ('E0001', 'P0001',
```

```
'2026-06-01', 10), ('E0001', 'SCCS', '2026-06-02', 15), ('E0002', 'SCCS', '2026-06-02', 40), ('E0002',
'P0001', '2026-06-03', 25), ('E0003', 'P0003', '2026-06-01', 35), ('E0004', 'P0003', '2026-06-02', 20),
('E0005', 'P0005', '2026-06-01', 12), ('E0006', 'P0005', '2026-06-02', 38), ('E0007', 'P0005',
'2026-06-03', 30), ('E0009', 'SCCS', '2026-06-04', 18));
Query OK, 10 rows affected (0.005 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

Question 7.25

```
MariaDB [work]> CREATE VIEW FemaleManagedProjects AS SELECT p.projNo, p.projName, p.deptNo FROM Project p
JOIN Department d ON p.deptNo = d.deptNo JOIN Employee e ON d.mgrEmpNo = e.empNo WHERE e.sex = 'F' ORDER
BY p.projNo;
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [work]> select * from FemaleManagedProjects;
```

projNo	projName	deptNo
P0001	Cloud Migration	D001
P0003	Q3 Strategic Mapping	D002
P0005	AI Predictive Engine	D005
SCCS	Secure Core Control Sys	D001

4 rows in set (0.006 sec)

Question 7.26

```
MariaDB [work]> CREATE VIEW EmpProjectDetails AS SELECT e.empNo, e.fName, e.lName, p.projName,
w.hoursWorked FROM WorksOn w JOIN Employee e ON w.empNo = e.empNo JOIN Project p ON w.projNo = p.projNo;
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [work]> select * from EmpProjectDetails;
```

empNo	fName	lName	projName	hoursWorked
E0001	Alice	Smith	Cloud Migration	10
E0001	Alice	Smith	Secure Core Control Sys	15
E0002	Bob	Jones	Cloud Migration	25
E0002	Bob	Jones	Secure Core Control Sys	40
E0003	Carol	Taylor	Q3 Strategic Mapping	35
E0004	David	Brown	Q3 Strategic Mapping	20
E0005	Emma	Davis	AI Predictive Engine	12
E0006	Frank	Miller	AI Predictive Engine	38
E0007	Grace	Wilson	AI Predictive Engine	30
E0009	Ivy	Thomas	Secure Core Control Sys	18

10 rows in set (0.002 sec)

Question 7.27

```
MariaDB [work]> CREATE VIEW EmpProject(empNo, projNo, totalHours) AS SELECT w.empNo, w.projNo,
SUM(hoursWorked) FROM Employee e, Project p, WorksOn w WHERE e.empNo = w.empNo AND p.projNo = w.projNo
GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.035 sec)
```

(a)

```
MariaDB [work]> SELECT* FROM EmpProject;
```

empNo	projNo	totalHours
E0001	P0001	10
E0001	SCCS	15
E0002	P0001	25
E0002	SCCS	40
E0003	P0003	35
E0004	P0003	20
E0005	P0005	12
E0006	P0005	38
E0007	P0005	30
E0009	SCCS	18

10 rows in set (0.031 sec)

(b)

```
MariaDB [work]> SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';
+-----+
```

```

| projNo |
+-----+
| SCCS   |
| SCCS   |
| SCCS   |
+-----+
3 rows in set (0.007 sec)

```

(c)

```

MariaDB [work]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';
+-----+
| COUNT(projNo) |
+-----+
|              0 |
+-----+
1 row in set (0.002 sec)
MariaDB [work]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E0001';
+-----+
| COUNT(projNo) |
+-----+
|              2 |
+-----+
1 row in set (0.002 sec)

```

(d)

```

MariaDB [work]> SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E0001 |          10 |
| E0002 |          25 |
| E0003 |          35 |
| E0004 |          20 |
| E0005 |          12 |
| E0006 |          38 |
| E0007 |          30 |
| E0009 |          18 |
+-----+-----+
8 rows in set (0.002 sec)

```

-- Analysis

- (a) `SELECT * FROM EmpProject;`
 Displays a summary of the total cumulative hours worked by each unique employee on each unique project.
- (b) `SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';`
 Returns a list of project numbers (all reading 'SCCS') for every employee assigned to it.
- (c) `SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';`
 Counts the total number of distinct projects that employee 'E1' has logged hours towards.
- (d) `SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;`
 Returns the total overall hours an employee worked across all projects.

Question 7.28

-- How would we maintain this as a materialized view?
 To keep 'ExpensiveParts' up to date when rows in 'Part' are inserted, updated, or deleted, we have two strategies we can use. First is to set up triggers on 'Part' that run and update the materialized view the moment the rows change.

-- How do we maintain the view without having to access the base table?
 To maintain the view without scanning the entire underlying Part table from scratch, we need to use an Incremental Refresh where instead of reading the whole table, we only look at the specific rows that were just modified.

-- Queries

-- The base table
 MariaDB [hotel]> CREATE TABLE Part (partNo VARCHAR(10), contract VARCHAR(10), partCost DECIMAL(10, 2), PRIMARY KEY (partNo, contract));
 Query OK, 0 rows affected (0.076 sec)

-- the materialized view table
 MariaDB [hotel]> CREATE TABLE MV_ExpensiveParts (partNo VARCHAR(10) PRIMARY KEY, qualifying_count INT DEFAULT 1);
 Query OK, 0 rows affected (0.038 sec)

```
-- Trigger that Handles inserts
MariaDB [hotel]> DELIMITER //
MariaDB [hotel]> CREATE TRIGGER trg_Part_Insert
-> AFTER INSERT ON Part
-> FOR EACH ROW
-> BEGIN
->     IF NEW.partCost > 1000 THEN
->         INSERT INTO MV_ExpensiveParts (partNo, qualifying_count)
->         VALUES (NEW.partNo, 1)
->         ON DUPLICATE KEY UPDATE qualifying_count = qualifying_count + 1;
->     END IF;
-> END //
```

Query OK, 0 rows affected (0.060 sec)

```
-- Trigger that Handles deletes
MariaDB [hotel]> DELIMITER //
MariaDB [hotel]> CREATE TRIGGER trg_Part_Delete
-> AFTER DELETE ON Part
-> FOR EACH ROW
-> BEGIN
->     IF OLD.partCost > 1000 THEN
->         -- Decrement the count for this part
->         UPDATE MV_ExpensiveParts
->         SET qualifying_count = qualifying_count - 1
->         WHERE partNo = OLD.partNo;
->
->         -- Clean up the row if no contracts price it > 1000 anymore
->         DELETE FROM MV_ExpensiveParts WHERE qualifying_count <= 0;
->     END IF;
-> END //
```

Query OK, 0 rows affected (0.065 sec)

-- Let's test to prove if it works

```
-- 1. Insert an inexpensive part (Should NOT appear in the MV)
MariaDB [hotel]> INSERT INTO Part VALUES ('P1', 'CNT-A', 500.00);
Query OK, 1 row affected (0.049 sec)
```

```
-- 2. Insert an expensive part under Contract A (Should appear in MV with count = 1)
MariaDB [hotel]> INSERT INTO Part VALUES ('P2', 'CNT-A', 1200.00);
Query OK, 1 row affected (0.004 sec)
```

```
-- 3. Insert the SAME part number under Contract B, also expensive (Count should become 2)
MariaDB [hotel]> INSERT INTO Part VALUES ('P2', 'CNT-B', 1500.00);
Query OK, 1 row affected (0.006 sec)
```

```
MariaDB [hotel]> SELECT * FROM MV_ExpensiveParts;
+-----+-----+
| partNo | qualifying_count |
+-----+-----+
| P2     |                2 |
+-----+-----+
1 row in set (0.035 sec)
```

-- let's test delete behavior (No Base Table Scan)

```
MariaDB [hotel]> DELETE FROM Part WHERE partNo = 'P2' AND contract = 'CNT-A';
Query OK, 1 row affected (0.043 sec)
```

```
MariaDB [hotel]> SELECT * FROM MV_ExpensiveParts;
+-----+-----+
| partNo | qualifying_count |
+-----+-----+
| P2     |                1 |
+-----+-----+
1 row in set (0.002 sec)
```